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Extra Worksheet – Lectures 11 and 12

COMP3670 / COMP6670: Introduction to Machine Learning

Throughout, the **running example** of the lectures is the sample of four observations (x_i, y_i):

$$(1, 2), \qquad (2, 5), \qquad (3, 4), \qquad (4, 9).$$

Question 1 The running example, from the data to the derivative

Consider the line through the origin, $h_\theta(x) = \theta x$, fitted to the running example under the squared loss, so that

$$E_{\text{in}}(\theta) = \frac{1}{4}\sum_{i=1}^{4} \big(y_i - \theta x_i\big)^2.$$

(a) Expand the sum and show that $E_{\text{in}}(\theta) = 7.5\,\theta^2 - 30\,\theta + 31.5$.

[*Hint*: expand the square and compute $\sum_i x_i^2$, $\sum_i x_i y_i$ and $\sum_i y_i^2$ first.]

(b) Compute the average rate of change of E_{in} when θ goes from 0 to 1, from 1 to 2, from 2 to 3, from 3 to 4 and from 0 to 4. Why is the last one equal to 0, even though the error changes a lot between $\theta = 0$ and $\theta = 4$?

(c) Show that the average rate of change between θ and $\theta + \Delta\theta$ is $15\,\theta - 30 + 7.5\,\Delta\theta$, and deduce $E'_{\text{in}}(\theta)$ from the definition of the derivative.

(d) At $\theta = 1$, use the linear approximation $E_{\text{in}}(\theta+\Delta\theta) \approx E_{\text{in}}(\theta)+E'_{\text{in}}(\theta)\,\Delta\theta$ to predict $E_{\text{in}}(1.1)$ and $E_{\text{in}}(2)$, and compare with the true values. Show that, for this function, the error of the approximation is *exactly* $7.5\,(\Delta\theta)^2$, whatever the value of θ.

(e) Compute $E'_{\text{in}}(\theta)$ a third way, with the "residual times input" formula

$$E'_{\text{in}}(\theta) = -\frac{2}{N}\sum_i x_i\,(y_i - \theta x_i),$$

writing the four terms of the sum explicitly.

Question 2 **Derivatives: the definition and the rules**

(a) Using the definition $f'(x) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$, exactly as was done for x^2 in the lecture, compute the derivatives of

$$\text{(i) } f(x) = x^3, \qquad \text{(ii) } f(x) = \frac{1}{x} \ (x \neq 0), \qquad \text{(iii) } f(x) = \sqrt{x} \ (x > 0).$$

[*Hint for* (iii): multiply the numerator and the denominator by $\sqrt{x + \Delta x} + \sqrt{x}$.]

(b) Write down the details of the chain rule for each line of the table of Lecture 11: name the inside function u, differentiate the outside with respect to u, and multiply by the derivative of the inside.

$$\text{(i) } (x^3 - 1)^{100}, \qquad \text{(ii) } e^{-x^2}, \qquad \text{(iii) } \log(1 - \theta), \qquad \text{(iv) } e^{-z}.$$

(c) Use the product rule to differentiate $x^2 e^x$, $x \log x$ and $p\,(1 - p)$.

(d) Compute the second derivative of $\sqrt{x}$, and check that it is negative for every $x > 0$.

(e) A quotient needs no rule of its own. Write $\frac{x}{1+x^2}$ as $x \cdot (1 + x^2)^{-1}$ and differentiate it with the product and chain rules.

Question 3 **The logistic function**

Let $\sigma(z) = \dfrac{1}{1 + e^{-z}}$ be the logistic function of Lecture 4.

(a) Writing $\sigma(z) = u^{-1}$ with $u = 1 + e^{-z}$, use the chain rule to show that $\sigma'(z) = \dfrac{e^{-z}}{(1 + e^{-z})^2}$.

(b) Show that this reduces to $\sigma'(z) = \sigma(z)\big(1 - \sigma(z)\big)$.

(c) Show that $\sigma'(z) \leq \frac{1}{4}$ for every z, with equality only at $z = 0$. [*Hint*: write $p = \sigma(z)$ and complete the square in $p(1 - p)$.]

(d) Show that $\sigma''(z) = \sigma(z)\big(1 - \sigma(z)\big)\big(1 - 2\sigma(z)\big)$, and deduce that σ is neither convex nor concave.

(e) Logistic regression on one observation (x, y), with a single weight w and no intercept, is the pipeline

$$w \ \longmapsto \ z = wx \ \longmapsto \ p = \sigma(z) \ \longmapsto \ \ell,$$

with $\ell = -\log p$ if $y = 1$, and $\ell = -\log(1 - p)$ if $y = 0$. Compute $\frac{d\ell}{dw}$ in both cases, as the product of the three local derivatives, and show that both answers are given by the single formula $\frac{d\ell}{dw} = (p - y)\, x$.

Question 4 **Critical points, the tests and convexity**

(a) The functions x^3, x^4 and $-x^4$ all have a critical point at $x = 0$. Show that the second derivative test says nothing about any of them, and use the *first* derivative test (the sign of f' on each side) to classify $x = 0$ in each case.

(b) Using the second derivative, decide whether each of the following functions is convex, concave or neither on its domain:

$$x^2, \quad e^x, \quad -\log x, \quad \log x, \quad \sqrt{x}, \quad -x^2, \quad x^3.$$

(c) The functions $|x|$ and the hinge $\max\{0,\, 1-m\}$ have a corner, so their second derivative does not exist everywhere. Prove that they are convex directly from the definition: $f(ax+by) \le af(x) + bf(y)$ for all x, y and all $a, b \ge 0$ with $a + b = 1$. [*Hint*: for $|x|$, use the triangle inequality $|u+v| \le |u|+|v|$. For the hinge, show that the maximum of two convex functions is convex.]

(d) Prove the three rules of Lecture 11, directly from the definition: if f and g are convex then (i) $f + g$ is convex, (ii) $c\,f$ is convex for every $c \ge 0$, and (iii) $\theta \mapsto f(s\,\theta + r)$ is convex for all $s, r \in \mathbb{R}$.

(e) Use (b) and (d) to show that $E_{\text{in}}(\theta) = \frac{1}{N}\sum_i (y_i - \theta x_i)^2$ is convex, whatever the data.

Question 5 **Critical points from a formula**

Let

$$f(x) = \frac{x^5}{5} - x^4 + \frac{2x^3}{3} + 2x^2 - 3x.$$

(a) Compute f', check that $f'(1) = 0$, and show that $f'(x) = (x-1)^2(x-3)(x+1)$.

(b) Find all the critical points of f, and classify each as a local minimum, a local maximum or neither, with the first derivative test.

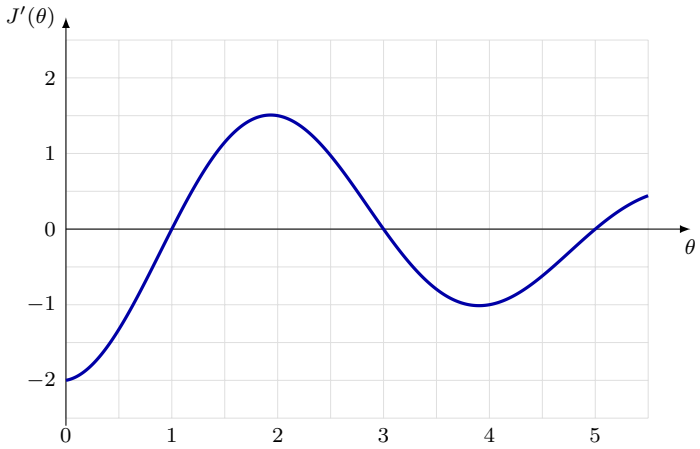
(c) Compute f'' at the critical points. At which critical point does the second derivative test fail, and why does the first derivative test still work there?

(d) Does f have a global minimiser on $\mathbb{R}$? On $[-1, \infty)$?

(e) Gradient descent with a small learning rate is started at $x(0) = 0$, at $x(0) = 2$ and at $x(0) = -2$. Where does each run go? Which runs end at a local minimum?

Question 6 **Reading a function from its derivative**

The plot shows the derivative $J'(\theta)$ of an error $J(\theta)$ on $[0, 5.5]$, and not the error itself. J' is itself differentiable.



(a) Find the critical points of J in $(0, 5.5)$ and classify each as a local minimum, a local maximum or neither.

(b) On which intervals is J increasing, and on which is it decreasing?

(c) On which intervals is J convex, and on which is it concave? (J is convex on an interval if the definition of convexity holds for any two points of that interval.) The derivative is lowest at $\theta = 0$ and has a local minimum near $\theta = 4$: are these minima of J?

(d) Gradient descent $\theta(t + 1) = \theta(t) - \alpha J'(\theta(t))$ with a small $\alpha > 0$ is started at $\theta(0) = 0.5$, 2.5, 3.5 and 5.5. Where does each run converge?

(e) Now take a larger learning rate, $\alpha = 1$, and start at $\theta(0) = 0$. Read the values of J' you need from the plot and compute $\theta(1)$ and $\theta(2)$. Near $\theta = 1$ the slope of J' is $J''(1) \approx 2.75$, so that $J'(\theta) \approx 2.75\,(\theta - 1)$. Use this approximation to explain why the iterates cannot settle at 1 when $\alpha = 1$, and find the learning rates for which gradient descent converges to 1 once it is close to it.

Question 7 **Maximum likelihood**

(a) For the coin of Lecture 4, with 7 heads in 10 tosses, $-\mathrm{LL}(\theta) = -\log 120 - 7\log\theta - 3\log(1-\theta)$ for $0 < \theta < 1$. Differentiate term by term, writing out the chain rule for the last term, solve $(-\mathrm{LL})'(\theta) = 0$, and use the second derivative to check that the solution is a minimiser of $-\mathrm{LL}$.

(b) Repeat (a) with k successes in N trials, $0 < k < N$, and show that $\hat{\theta} = k/N$. Show that the minimiser is global. What happens when $k = 0$?

(c) Let $x_1, \ldots, x_N$ be an i.i.d. sample of the Normal distribution with mean μ and variance v, whose density is $\frac{1}{\sqrt{2\pi v}} \exp\big(-\frac{(x-\mu)^2}{2v}\big)$. Assume that the x_i are not all equal.

(i) Show that $-\mathrm{LL}(\mu, v) = \frac{N}{2}\log(2\pi) + \frac{N}{2}\log v + \frac{1}{2v}\sum_{i=1}^{N}(x_i - \mu)^2$.

(ii) Compute the two partial derivatives of $-\mathrm{LL}$, set them to zero, and show that the only critical point is

$$\hat{\mu} = \bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad \hat{v} = \frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2.$$

(iii) Show that this critical point is the global minimiser of $-\mathrm{LL}$. [*Hint*: first show that, for every fixed v, the best μ is $\bar{x}$. Then study the function of v alone, $g(v) = -\mathrm{LL}(\bar{x}, v)$, with the first derivative test.]

Question 8 **Matrix calculus and the residual sum of squares**

Let $X \in \mathbb{R}^{N\times m}$ be a data matrix, $\mathbf{Y} \in \mathbb{R}^N$ and $\mathrm{RSS}(\boldsymbol{\theta}) = \|\mathbf{Y} - X\boldsymbol{\theta}\|^2$.

(a) Writing the functions coordinate by coordinate, show that $\nabla(\|\mathbf{x}\|^2) = 2\mathbf{x}$ and that, for any $m \times m$ matrix A, $\nabla(\mathbf{x}^T A\mathbf{x}) = (A + A^T)\mathbf{x}$. Deduce the formula $2A\mathbf{x}$ of the lecture when A is symmetric.

(b) Show that $\mathrm{RSS}(\boldsymbol{\theta}) = \boldsymbol{\theta}^T X^T X\boldsymbol{\theta} - 2\,\mathbf{Y}^T X\boldsymbol{\theta} + \mathbf{Y}^T\mathbf{Y}$. Explain why the two terms $\mathbf{Y}^T X\boldsymbol{\theta}$ and $\boldsymbol{\theta}^T X^T\mathbf{Y}$ of the expansion are equal. Deduce $\nabla\mathrm{RSS}(\boldsymbol{\theta}) = -2X^T(\mathbf{Y} - X\boldsymbol{\theta})$.

(c) For the running example with an intercept, write X and $\mathbf{Y}$, compute $X^T X$ and $X^T\mathbf{Y}$, and use $\nabla\mathrm{RSS}$ to compute the gradient of the in-sample error $E_{\mathrm{in}}(\theta_0, \theta_1) = \frac{1}{4}\sum_i (y_i - \theta_0 - \theta_1 x_i)^2$ of the line $h_{\boldsymbol{\theta}}(x) = \theta_0 + \theta_1 x$.

(d) Show that $\mathbf{v}^T(2X^T X)\mathbf{v} = 2\|X\mathbf{v}\|^2$ for every $\mathbf{v} \in \mathbb{R}^m$. What does it say about the convexity of the RSS?

(e) Let $\hat{\boldsymbol{\theta}}$ satisfy $X^T(\mathbf{Y} - X\hat{\boldsymbol{\theta}}) = \mathbf{0}$. Show that $\mathrm{RSS}(\hat{\boldsymbol{\theta}} + \mathbf{v}) = \mathrm{RSS}(\hat{\boldsymbol{\theta}}) + \|X\mathbf{v}\|^2$ for every $\mathbf{v}$.

(f) Suppose the four inputs of the running example had all been equal to $x_i = 2.5$, with the same y_i. Find a vector $\mathbf{v} \neq \mathbf{0}$ with $X\mathbf{v} = \mathbf{0}$, describe the set of all minimisers (θ_0, θ_1), and show that they are exactly the lines through the point $(\bar{x}, \bar{y}) = (2.5, 5)$.

Question 9 **Ridge regression**

Let $J(\boldsymbol{\theta}) = \|\mathbf{Y} - X\boldsymbol{\theta}\|^2 + \lambda\|\boldsymbol{\theta}\|^2$ with $\lambda > 0$.

(a) Compute ∇J and show that its critical points solve $(X^TX + \lambda I)\boldsymbol{\theta} = X^T\mathbf{Y}$.

(b) Show that if $X^TX\mathbf{v} = \gamma\mathbf{v}$ then $(X^TX + \lambda I)\mathbf{v} = (\gamma + \lambda)\mathbf{v}$. Show also that every eigenvalue γ of X^TX satisfies $\gamma \geq 0$. Deduce that $X^TX + \lambda I$ is invertible and that J is strictly convex.

(c) For the line through the origin, $h_\theta(x) = \theta x$, show that $\hat{\theta}^{\text{RIDGE}} = \frac{\sum_i x_i y_i}{\sum_i x_i^2 + \lambda}$. Compute it on the running example for $\lambda = 0$ (ordinary least squares), 10 and 30, and describe what happens as $\lambda \to \infty$.

(d) A data set contains the same feature twice: X has two identical columns, both equal to $(1, 2, 3, 4)^T$ (no intercept), and $\mathbf{Y} = (2, 5, 4, 9)^T$.

(i) Show that X^TX is not invertible, and that every (θ_1, θ_2) with $\theta_1 + \theta_2 = 2$ minimises the RSS.

(ii) Solve the ridge equations and show that their unique solution is $\theta_1 = \theta_2 = \frac{60}{60+\lambda}$. Which of the least-squares solutions does it approach as $\lambda \to 0$?

(e) Derive the gradient descent step for ridge regression, $\boldsymbol{\theta} \leftarrow (1 - 2\alpha\lambda)\,\boldsymbol{\theta} + 2\alpha\, X^T(\mathbf{Y} - X\boldsymbol{\theta})$.

Question 10 **The gradient of the cross-entropy**

Logistic regression with weights $\mathbf{w} \in \mathbb{R}^{d+1}$, where each input $\mathbf{x}$ has a 1 in its first coordinate, has loss $\ell(\mathbf{w}) = -\,y \log \sigma(\mathbf{w}^T\mathbf{x}) - (1-y) \log\big(1 - \sigma(\mathbf{w}^T\mathbf{x})\big)$ on one observation $(\mathbf{x}, y)$, with $y \in \{0, 1\}$.

(a) Writing $z = \mathbf{w}^T\mathbf{x}$, compute $\frac{d\ell}{dz}$ and $\frac{\partial z}{\partial w_j}$, and deduce that $\nabla\ell(\mathbf{w}) = \big(\sigma(\mathbf{w}^T\mathbf{x}) - y\big)\,\mathbf{x}$.

(b) Show that the Hessian of ℓ is $\sigma(z)\big(1 - \sigma(z)\big)\,\mathbf{x}\mathbf{x}^T$, and that it is positive semidefinite. Conclude that the average cross-entropy $J(\mathbf{w}) = \frac{1}{N}\sum_i \ell_i(\mathbf{w})$ is convex.

(c) Take one feature, so $\mathbf{x} = (1, x)^T$ and $\mathbf{w} = (w_0, w_1)^T$, and the four observations

$$(x, y) \;=\; (-1, 0), \quad (1, 1), \quad (2, 1), \quad (0, 0).$$

Starting from $\mathbf{w} = (0, 0)^T$, compute ∇J, take one step of gradient descent with $\alpha = 1$, and compare J before and after the step.

Question 11 **Gradient descent by hand**

Let $J(\theta) = 7.5\,\theta^2 - 30\,\theta + 31.5 = 7.5\,(\theta - 2)^2 + 1.5$, the running example of Lecture 12, and run gradient descent $\theta(t+1) = \theta(t) - \alpha J'(\theta(t))$ from $\theta(0) = 0$.

(a) With $\alpha = 0.04$, reproduce the table of the lecture for $t = 0, 1, 2, 3, 4$: $\theta(t)$, $J'(\theta(t))$ and $J(\theta(t))$.

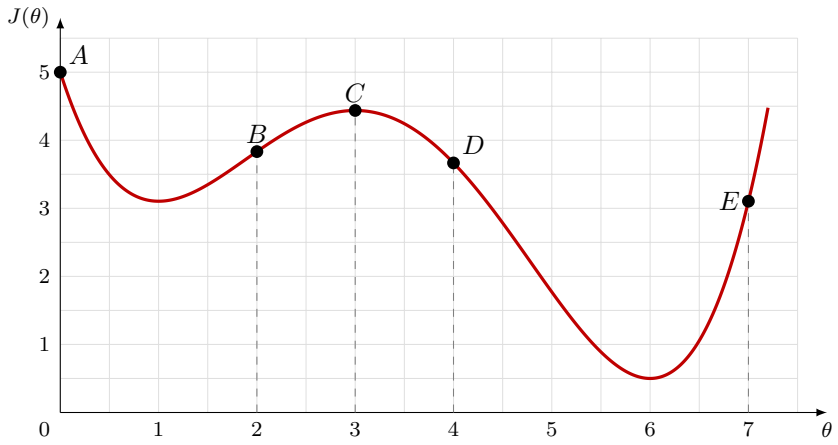
(b) Show that $\theta(t+1) - 2 = (1 - 15\alpha)\big(\theta(t) - 2\big)$ for every t, and deduce the formula $\theta(t) = 2 - 2(1 - 15\alpha)^t$. How far from the minimiser is the iterate after 10 steps with $\alpha = 0.04$?

(c) Show that the iterates converge to 2 if and only if $0 < \alpha < 2/15$. Use the formula of (b) to compute the iterates of the lecture for $\alpha = 0.005$ (after 8 steps) and for $\alpha = 0.14$ (the first three). What happens for $\alpha = 2/15$ and for $\alpha = 1/15$?

(d) Show that $J(\theta(t)) - J(2) = 30\,(0.16)^t$ when $\alpha = 0.04$. Compare it with the bound of the theorem of Lecture 12, $\frac{\|\theta(0) - \hat{\theta}\|^2}{2\alpha t}$, for $t = 1$ and $t = 5$, after checking that the theorem applies.

Question 12 **One step from five starting points**

The plot shows an error $J(\theta)$ and five marked values A, B, C, D, E of the parameter.



(a) One step of gradient descent, $\theta \leftarrow \theta - \alpha J'(\theta)$, is taken from each of the five values, with the same small $\alpha > 0$. For each one, say whether the parameter moves left, moves right or stays where it is, and rank the five starting points by the length of the step.

(b) From one of the five values gradient descent never moves. Is that value a minimiser of J? What would happen if the parameter started a tiny distance to its left, or to its right?

(c) With a small α, where does gradient descent converge from each of the five values? Which runs find the global minimiser?

(d) The error is $J(\theta) = \frac{\theta^4}{16} - \frac{5\theta^3}{6} + \frac{27\theta^2}{8} - \frac{9\theta}{2} + 5$. Show that $J'(\theta) = \frac{1}{4}(\theta - 1)(\theta - 3)(\theta - 6)$ and compute the slope of J at the five marked values. Do they confirm your ranking in (a)?

(e) Take one step with the large learning rate $\alpha = 1$ from each of the five values. Which steps jump into the other valley? Compare J before and after the step from E, and explain why this does not contradict the fact that gradient descent decreases J.

Question 13 Fast convergence when the curvature is bounded below

Let $J : \mathbb{R} \to \mathbb{R}$ be differentiable, with a minimiser $\hat{\theta}$, so that $J'(\hat{\theta}) = 0$. Assume that there are constants $0 < \mu \le K$ such that

$$\text{(B)} \qquad \mu \;\le\; \frac{J'(x) - J'(y)}{x - y} \;\le\; K \qquad \text{for all } x \neq y.$$

(For a twice differentiable J this says $\mu \le J''(x) \le K$ everywhere: the curvature is bounded above *and below*.) Gradient descent is run as $\theta(t+1) = \theta(t) - \alpha J'(\theta(t))$ with $0 < \alpha \le 1/K$. You may use without proof the inequality

$$\text{(Q)} \qquad J(y) \;\le\; J(x) + J'(x)\,(y - x) + \frac{K}{2}\,(y - x)^2 \qquad \text{for all } x, y,$$

which follows from the upper bound in (B).

(a) Suppose $\theta(t) \neq \hat{\theta}$ and let $c_t = \frac{J'(\theta(t))}{\theta(t) - \hat{\theta}}$. Show that $\mu \le c_t \le K$ and that $\theta(t+1) - \hat{\theta} = (1 - \alpha c_t)(\theta(t) - \hat{\theta})$.

(b) Show that $0 \le 1 - \alpha c_t \le 1 - \alpha\mu$, and deduce that

$$|\theta(T) - \hat{\theta}| \;\le\; (1 - \alpha\mu)^T\,|\theta(0) - \hat{\theta}| \qquad \text{for every } T \ge 0.$$

(Treat separately the case where some iterate equals $\hat{\theta}$.)

(c) Use (Q) with $x = \hat{\theta}$ and $y = \theta(T)$ to show that

$$J(\theta(T)) - J(\hat{\theta}) \;\le\; \frac{K}{2}\,(1 - \alpha\mu)^{2T}\,(\theta(0) - \hat{\theta})^2.$$

(d) Show that J has only one minimiser.

(e) For $J(\theta) = 7.5\,\theta^2 - 30\,\theta + 31.5$, find the best constants μ and K. With $\alpha = 0.04$ and $\theta(0) = 0$, how many steps T does (c) require to guarantee $J(\theta(T)) - J(\hat{\theta}) \le 10^{-6}$? How many does the bound $\frac{(\theta(0) - \hat{\theta})^2}{2\alpha T}$ of Lecture 12 require for the same guarantee?

Question 14 **Stochastic gradient descent on the running example**

Take the one-parameter running example, $J(\theta) = \frac{1}{4}\sum_{i=1}^{4} \ell_i(\theta)$ with $\ell_i(\theta) = (y_i - \theta x_i)^2$.

(a) A mini-batch B of b distinct indices is drawn uniformly at random among all subsets of size b of $\{1, \dots, N\}$. Show that each index belongs to B with probability b/N, and deduce that $\mathbf{g} = \frac{1}{b}\sum_{i\in B} \nabla\ell_i(\theta)$ is an unbiased estimate of $\nabla J(\theta)$. [*Hint*: write $\mathbf{g} = \frac{1}{b}\sum_{i=1}^{N} \mathbb{1}\{i \in B\}\, \nabla\ell_i(\theta)$.]

(b) Compute the slope y_i/x_i that each observation would choose on its own, and the four individual derivatives $\ell_i'(2)$ at the minimiser $\hat{\theta} = 2$. Check that their average is 0.

(c) SGD with batches of size 1 and $\alpha = 0.02$ is at $\theta = 2$. Where is it after one more step, for each of the four possible indices?

(d) At $\theta = 2$, compute the mean and the variance of the single-observation gradient $\ell_I'(2)$, where I is uniform on $\{1, 2, 3, 4\}$. What is the variance of the average of b such gradients, drawn independently (that is, with replacement)?

Question 15 **Momentum, Adagrad and Adam by hand**

Use again $J(\theta) = 7.5\,\theta^2 - 30\,\theta + 31.5$, with $J'(\theta) = 15\,\theta - 30$. All methods start at $\theta(1) = 0$ with $m(0) = s(0) = 0$, and run for $t = 1, 2, \dots$ with the updates of Lecture 12 (with θ in place of $\mathbf{w}$, and $\epsilon = \epsilon_{\text{root}} = 0$).

(a) **Heavy ball**, with $\alpha = 0.04$ and $\beta = 0.9$: compute $\theta(2)$, $\theta(3)$ and $\theta(4)$, and compare with gradient descent with the same α after the same number of steps (Question 11, where the start is called $\theta(0)$). Is momentum helping on this problem?

(b) Show that the heavy-ball momentum is $m(t) = -\alpha\sum_{r=1}^{t} \beta^{t-r} J'(\theta(r))$. If the gradient were constant, $J'(\theta(r)) = g$ for all r, what would the step $m(t)$ become for large t? Compare with the gradient descent step when $\beta = 0.9$.

(c) **Adagrad**, with $\alpha = 1$: compute $\theta(2)$ and $\theta(3)$. Show that, if $J'(\theta(1)) \neq 0$, the first Adagrad step has length exactly α, whatever the gradient.

(d) **Adam**, with $\alpha = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.999$: show that, if $J'(\theta(1)) \neq 0$, the first step is $\theta(2) = \theta(1) - \alpha\ \text{sign}\big(J'(\theta(1))\big)$, and compute $\theta(2)$ and $\theta(3)$.